Fiscal Year:	FY 2013	**Task Last Updated:**	FY 08/30/2013
PI Name:	Salas, Eduardo Ph.D.		
Project Title:	Optimizing Crew Performance in Long Duration Space Exploration: Best Practices for Team Training and Cohesion Measurement		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **Team**:Risk of Performance and Behavioral Health Decrements Due to Inadequate Cooperation, Coordination, Communication, and Psychosocial Adaptation within a Team		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77005	**Congressional District:**	7
Comments:	NOTE: Previous affiliation was University of Central Florida, until mid-2015		
Project Type:	Ground	**Solicitation / Funding Source:**	2008 Crew Health NNJ08ZSA002N
Start Date:	08/15/2009	**End Date:**	12/31/2013
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	5
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Leveton, Lauren	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date is now 12/31/2013 per NSSC information (Ed., 7/29/13) NOTE: End date is now 8/14/2013 per NSSC information (Ed., 6/15/2012) NOTE: period of performance changed to 8/15/2009-8/14/2012 (from 5/22/09-5/21/12) per JSC (3/10)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Fiore, Stephen (University of Central Florida) Smith-Jentsch, Kimberly (University of Central Florida)		
Grant/Contract No.:	NNX09AK48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This project addresses questions regarding methods and technologies for training crews to maintain crew cohesion and optimal performance during exploration missions (BHP Team GAP5; IRP Gap - BHP 2.3.1) and metrics for monitoring crew cohesion (BHP Team GAP2; IRP Gap - BHP 2.2.1). Specific aims of this project are threefold: (1) identify evidence-based guidelines/best practices for training to maximize team cohesion, and team performance, (2) design, develop, and validate evidence-based instructional strategies to mitigate performance failures from cohesion decrements among spaceflight crews and coordinating ground crews, and (3) design, develop, and validate an evidence-based index measuring and diagnosing cohesion over the course of a mission. These specific project aims meet NASA goals and objectives (BHP Team Gap 2 and Team Gap5) by capturing cohesion levels shown to be integral to long duration spaceflight mission success as well as developing countermeasures designed to mitigate the negative impact of cohesion issues. Organized as a multi-year project, primary tasks for the first year included the development of recommendations for training crews to optimize cohesion and team performance, mitigate negative impacts of long-duration missions, and measuring crew cohesion over time. Second and third years of the project focus on applying recommendations derived from year one research by developing, implementing, and evaluating instructional strategies to maximize crew cohesion and mitigate negative psychosocial impacts of long-duration missions. Overarching project goals are to: (1) mitigate performance failures due to a lack of cohesion between spaceflight crews and coordinating ground crews, (2) diagnose cohesion decrements during exploration missions, and (3) provide just-in-time training to improve any noted cohesion decrements during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Working together on long-duration space exploration missions in conjunction with ground control requires the ability to communicate, coordinate, and cooperate for extended durations under complex, dynamic conditions such as extreme isolation and confinement (NASA, 2009). Factors related to team cohesion (e.g. interpersonal conflict, impaired communication) were noted as contributors in both the Challenger and Columbia shuttle accidents. Determining the best strategies for equipping crew members with the cognitive, behavioral, and attitudinal tools necessary to cope under such intense conditions while maintaining optimal performance, and developing a means to unobtrusively monitor crew adaptation over time are critical to the success of such missions. The multipronged research approach will yield (1) an understanding of the factors that contribute to a lack of cohesion and (2) the identification of stressors relevant to spaceflight crews, which will aid in pinpointing training strategies and methodologies to equip team members with the competencies necessary for coping with isolated, confined environments for long durations. Additionally, the focus on self-correction and regulation will enable teams to maintain levels of cohesion and team performance as well as to mitigate the negative impact of cohesion decrements over the course of long duration spaceflight.
Task Progress:	Year 4 (Y4) efforts have focused on: (1) data collection for a field study investigating the impact of debriefing conditions on team learning and performance, (2) laboratory experiments investigating team composition and strain and its relation to team performance, (3) the development of guidelines for measuring team cohesion to monitor long duration space flight (LDSF) teams, and (4) the development of team training guidelines and best practices framework. Regarding efforts within the three main thrusts provided in the initial proposal, efforts within Thrust 1 (Cohesion) were focused on (1) continued literature analysis for the development of cohesion measurement, (2) the development of team training guidelines and best practices, and (3) experimental testing investigating the factors associated with team cohesion. Efforts associated with Thrust 2 (Stress) involved: (1) developing guidelines for team level stress measurement, and (2) experimental testing investigating the factors contributing to team member strain. Efforts involved in Thrust 3 (Self-Regulation) primarily involved data collection for a field study investigating the validity of utilizing Team Dimensional Training (TDT) in a NASA Mission Control setting. TDT is a structured debriefing style for guiding teams to develop shared and accurate teamwork knowledge and facilitate their development of teamwork skills. Additionally, the self-regulation thrust involved data collection and analysis of multiple laboratory experiments investigating the role of team composition and team member strain on team performance, and analyses of archival analog data. Project Pegasus has received a no-cost extension until the end of the year 2013 to provide an opportunity for collecting performance data in the TDT field validation study. Y4 progress so far has included several core tasks including: Literature and archival data analysis (Task 3), the development of cohesion measurement guidelines for LDSF teams (Task 4), laboratory experimental investigations focusing on composition factors and its relation to team performance (Task 5), and data collection for the validation study of the TDT debriefing strategy (Task 9).
Bibliography Type:	Description: (Last Updated: 11/21/2025)
Abstracts for Journals and Proceedings	Fiore SM, Wiltshire TJ, Oglesby JM, O'Keefe WS, Salas E. "Macrocognition in Mission Control teams: Understanding complex collaborative problem solving processes." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. Presentation. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Abstracts for Journals and Proceedings	Salas E, Smith-Jentsch K, Fiore SM, Oglesby JM, Dietz A, Sierra MJ, Grossman R. "Optimizing crew performance in long duration space exploration: Best practices for team training." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Abstracts for Journals and Proceedings	Sierra MJ, Smith-Jentsch K, Joseph DL, Salas E. "Crew member personality fit and job strain: Exploring crew personality composition as a stress countermeasure for long-duration spaceflight." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Abstracts for Journals and Proceedings	Sierra MJ, Smith-Jentsch KA, Salas E. "To laugh or not to laugh: The potential impact of crew member humor on team process and performance during long-duration spaceflight." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013

Abstracts for Journals and Proceedings	Thayer AL, Smith-Jentsch KA, Salas E. "Team composition and coordination: An empirical examination in air traffic control teams." 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. 2013 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-14, 2013. , Feb-2013
Abstracts for Journals and Proceedings	Sierra MJ, Joseph DL, Smith-Jentsch K. "Neuroticism fit and stress: Can two wrongs make a right?" 2013 INGRoup Conference, Atlanta, GA, July 11-13, 2013. 2013 INGRoup Conference, Atlanta, GA, July 11-13, 2013. , Jul-2013
Abstracts for Journals and Proceedings	Dalal D, Joseph DL, Smith-Jentsch KA, Sierra MJ. "Leader-follower assertiveness fit: An asset or a handicap?" 2013 (28th annual conference) Meeting of the Society for Industrial-Organizational Psychology, Houston, TX, April 10-15, 2013. 2013 (28th annual conference) Meeting of the Society for Industrial-Organizational Psychology, Houston, TX, April 10-15, 2013. , Apr-2013
Abstracts for Journals and Proceedings	Smith-Jentsch KA, Sierra MJ, Salas E. "Countdown to Mars: Sequencing team training for the long haul." 2013 (28th annual conference) Meeting of the Society for Industrial-Organizational Psychology, Houston, TX, April 10-15, 2013. 2013 (28th annual conference) Meeting of the Society for Industrial-Organizational Psychology, Houston, TX, April 10-15, 2013. , Apr-2013
Abstracts for Journals and Proceedings	Sierra MJ, Rico R, Smith-Jentsch K, Mihm D. "Antecedents and outcomes of team situational models." 73rd Annual Meeting of the Academy of Management, Orlando, FL, August 9-13, 2013. 73rd Annual Meeting of the Academy of Management, Orlando, FL, August 9-13, 2013. , Aug-2013
Articles in Peer-reviewed Journals	Fiore SM, Wiltshire TJ, Oglesby JM, O'Keefe WS, Salas E. "Examining complex collaborative problem solving processes in Mission Control." Aviation, Space, and Environmental Medicine. Under review as of September 2013. , Sep-2013
Books/Book Chapters	Gregory ME, Feitosa J, Driskell T, Salas E, Vessey WB. "Designing, delivering, and evaluating team training in organizations: Principles that work. Chapter 14." in "Developing and Enhancing Teamwork in Organizations: Evidence-based Best Practices and Guidelines." Ed. E. Salas, S. I. Tannenbaum, D. Cohen, G. Latham. San Francisco, CA : Jossey-Bass, 2013., Jun-2013
Books/Book Chapters	Driskell T, Salas E, Vessey WB. "Team training in virtual environments: A dual approach." in "Handbook of virtual environments: Design, implementation, and applications. 2nd ed." Ed. K.M. Stanney, K.S. Hale. Boca Raton, Fla. : CRC Press, in press as of September 2013., Sep-2013
NASA Technical Documents	Fiore S, Wiltshire T, Oglesby J, Salas E. "University of Central Florida's contribution to NASA's complex and collaborative team problem solving technical interchange meeting (TIM): July 30-31, 2012." University of Central Florida's contribution to NASA's complex and collaborative team problem solving technical interchange meeting (TIM): July 30-31, 2012. July 30-31, 2012. , Jul-2012
Papers from Meeting Proceedings	Dietz AS, Sierra MJ, Smith-Jentsch K, Salas E. "Guiding principles for team stress measurement." 56th Annual Meeting of the Human Factors and Ergonomics Society, Boston, MA, October 22-26, 2012. Human Factors and Ergonomics Society (HFES) 56th Annual Meeting Proceedings. Santa Monica, CA: Human Factors and Ergonomics Society, 2012. p. 1074-1078. , Oct-2012
Papers from Meeting Proceedings	Oglesby J, Salas E. "The issue of monotony and low workload in spaceflight: Considerations for the mission to Mars." 56th Annual Meeting of the Human Factors and Ergonomics Society, Boston, MA, October 22-26, 2012. Human Factors and Ergonomics Society (HFES) 56th Annual Meeting Proceedings. Santa Monica, CA: Human Factors and Ergonomics Society, 2012. p. 1782-1786. , Oct-2012